

$$\begin{aligned}
& \frac{-2}{16\pi^2} \left(-\frac{2}{81} \sum_p g_1^4 LF_{3,0}[m_d^{-P}] \delta_{i1i2} + \frac{5}{108} \sum_p g_1^4 LF_{4,-1}[m_d^{-P}] \delta_{i1i2} - \frac{8}{495} \sum_p g_1^4 LF_{5,-2}[m_d^{-P}] \delta_{i1i2} - \right. \\
& \quad \frac{2}{27} \sum_p g_1^4 LF_{3,0}[m_e^{-P}] \delta_{i1i2} + \frac{5}{36} \sum_p g_1^4 LF_{4,-1}[m_e^{-P}] \delta_{i1i2} - \frac{8}{135} \sum_p g_1^4 LF_{5,-2}[m_e^{-P}] \delta_{i1i2} - \\
& \quad \frac{1}{27} \sum_p g_1^4 LF_{3,0}[m_l^{-P}] \delta_{i1i2} + \frac{5}{72} \sum_p g_1^4 LF_{4,-1}[m_l^{-P}] \delta_{i1i2} - \frac{4}{135} \sum_p g_1^4 LF_{5,-2}[m_l^{-P}] \delta_{i1i2} - \\
& \quad \frac{1}{81} \sum_p g_1^4 LF_{3,0}[m_q^{-P}] \delta_{i1i2} + \frac{5}{216} \sum_p g_1^4 LF_{4,-1}[m_q^{-P}] \delta_{i1i2} - \frac{4}{495} \sum_p g_1^4 LF_{5,-2}[m_q^{-P}] \delta_{i1i2} - \\
& \quad \frac{8}{81} \sum_p g_1^4 LF_{3,0}[m_u^{-P}] \delta_{i1i2} + \frac{5}{27} \sum_p g_1^4 LF_{4,-1}[m_u^{-P}] \delta_{i1i2} - \frac{32}{495} \sum_p g_1^4 LF_{5,-2}[m_u^{-P}] \delta_{i1i2} - \\
& \quad \frac{1}{27} g_1^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} - \frac{1}{18} g_1^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} + \frac{8}{135} g_1^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} - \\
& \quad \frac{1}{324} g_1^4 LF_{2,1,0}[m_1, m_d^{i1}] \delta_{i1i2} - \frac{1}{324} g_1^4 LF_{2,2,-1}[m_1, m_d^{i1}] \delta_{i1i2} + \\
& \quad \frac{1}{162} g_1^4 LF_{3,1,-1}[m_1, m_d^{i1}] \delta_{i1i2} - \frac{1}{324} g_1^4 LF_{4,1,-2}[m_1, m_d^{i1}] \delta_{i1i2} - \\
& \quad \frac{1}{324} g_1^4 LF_{2,1,0}[m_1, m_d^{i2}] \delta_{i1i2} - \frac{1}{324} g_1^4 LF_{2,2,-1}[m_1, m_d^{i2}] \delta_{i1i2} + \\
& \quad \frac{1}{162} g_1^4 LF_{3,1,-1}[m_1, m_d^{i2}] \delta_{i1i2} - \frac{1}{324} g_1^4 LF_{4,1,-2}[m_1, m_d^{i2}] \delta_{i1i2} - \\
& \quad \frac{1}{36} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{2,1,0}[m_1, m_q^P] + \frac{1}{18} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{3,1,-1}[m_1, m_q^P] - \\
& \quad \frac{1}{36} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{4,1,-2}[m_1, m_q^P] - \frac{3}{4} g_2^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{2,1,0}[m_2, m_q^P] + \\
& \quad \frac{3}{2} g_2^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{3,1,-1}[m_2, m_q^P] - \frac{3}{4} g_2^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{4,1,-2}[m_2, m_q^P] - \\
& \quad \frac{1}{27} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_d^{i1}] \delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_d^{i1}] \delta_{i1i2} + \\
& \quad \frac{2}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_d^{i1}] \delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_d^{i1}] \delta_{i1i2} - \\
& \quad \frac{1}{27} g_1^2 g_3^2 LF_{2,1,0}[m_3, m_d^{i2}] \delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2 LF_{2,2,-1}[m_3, m_d^{i2}] \delta_{i1i2} + \\
& \quad \frac{2}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_3, m_d^{i2}] \delta_{i1i2} - \frac{1}{27} g_1^2 g_3^2 LF_{4,1,-2}[m_3, m_d^{i2}] \delta_{i1i2} - \\
& \quad \frac{4}{3} g_3^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{2,1,0}[m_3, m_q^P] + \frac{8}{3} g_3^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{3,1,-1}[m_3, m_q^P] - \\
& \quad \frac{4}{3} g_3^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{4,1,-2}[m_3, m_q^P] - \frac{1}{9} g_1^2 \overline{a}_d^{pr} a_d^{pr} LF_{2,2,0}[m_d^r, m_q^P] \delta_{i1i2} + \\
& \quad \frac{1}{18} g_1^2 \overline{a}_d^{pr} a_d^{pr} LF_{3,2,-1}[m_d^r, m_q^P] \delta_{i1i2} + \frac{1}{162} g_1^4 LF_{2,1,0}[m_d^{i1}, m_1] \delta_{i1i2} - \\
& \quad \frac{1}{324} g_1^4 LF_{3,1,-1}[m_d^{i1}, m_1] \delta_{i1i2} + \frac{2}{27} g_1^2 g_3^2 LF_{2,1,0}[m_d^{i1}, m_3] \delta_{i1i2} - \\
& \quad \frac{1}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_d^{i1}, m_3] \delta_{i1i2} + \frac{1}{162} g_1^4 LF_{2,1,0}[m_d^{i2}, m_1] \delta_{i1i2} - \\
& \quad \frac{1}{324} g_1^4 LF_{3,1,-1}[m_d^{i2}, m_1] \delta_{i1i2} + \frac{2}{27} g_1^2 g_3^2 LF_{2,1,0}[m_d^{i2}, m_3] \delta_{i1i2} - \\
& \quad \frac{1}{27} g_1^2 g_3^2 LF_{3,1,-1}[m_d^{i2}, m_3] \delta_{i1i2} - \frac{1}{9} g_1^2 \overline{a}_e^{pr} a_e^{pr} LF_{2,2,0}[m_e^r, m_l^P] \delta_{i1i2} + \\
& \quad \frac{1}{18} g_1^2 \overline{a}_e^{pr} a_e^{pr} LF_{3,2,-1}[m_e^r, m_l^P] \delta_{i1i2} + \frac{1}{9} g_1^2 \overline{a}_e^{pr} a_e^{pr} LF_{3,1,0}[m_l^P, m_e^r] \delta_{i1i2} + \\
& \quad \frac{1}{9} g_1^2 \overline{a}_e^{pr} a_e^{pr} LF_{3,2,-1}[m_l^P, m_e^r] \delta_{i1i2} - \frac{1}{4} g_1^2 \overline{a}_e^{pr} a_e^{pr} LF_{4,1,-1}[m_l^P, m_e^r] \delta_{i1i2} + \\
& \quad \frac{1}{9} g_1^2 \overline{a}_e^{pr} a_e^{pr} LF_{5,1,-2}[m_l^P, m_e^r] \delta_{i1i2} + \frac{1}{9} g_1^2 \overline{a}_d^{pr} a_d^{pr} LF_{3,1,0}[m_q^P, m_d^r] \delta_{i1i2} + \\
& \quad \frac{1}{9} g_1^2 \overline{a}_d^{pr} a_d^{pr} LF_{3,2,-1}[m_q^P, m_d^r] \delta_{i1i2} - \frac{5}{12} g_1^2 \overline{a}_d^{pr} a_d^{pr} LF_{4,1,-1}[m_q^P, m_d^r] \delta_{i1i2} + \\
& \quad \frac{1}{3} g_1^2 \overline{a}_d^{pr} a_d^{pr} LF_{5,1,-2}[m_q^P, m_d^r] \delta_{i1i2} + \frac{1}{18} g_1^2 \tilde{\mu}^2 \overline{y}_u^{pr} y_u^{pr} LF_{2,2,0}[m_q^P, m_u^r] \delta_{i1i2} - \\
& \quad \frac{1}{36} g_1^2 \tilde{\mu}^2 \overline{y}_u^{pr} y_u^{pr} LF_{3,2,-1}[m_q^P, m_u^r] \delta_{i1i2} - \frac{1}{18} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{2,1,0}[m_q^P, \tilde{\mu}] + \\
& \quad \frac{1}{36} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{2,2,-1}[m_q^P, \tilde{\mu}] + \frac{1}{36} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{3,1,-1}[m_q^P, \tilde{\mu}] - \\
& \quad \frac{1}{18} g_1^2 \tilde{\mu}^2 \overline{y}_u^{pr} y_u^{pr} LF_{3,1,0}[m_u^r, m_q^P] \delta_{i1i2} - \frac{1}{18} g_1^2 \tilde{\mu}^2 \overline{y}_u^{pr} y_u^{pr} LF_{3,2,-1}[m_u^r, m_q^P] \delta_{i1i2} - \\
& \quad \frac{1}{6} g_1^2 \tilde{\mu}^2 \overline{y}_u^{pr} y_u^{pr} LF_{4,1,-1}[m_u^r, m_q^P] \delta_{i1i2} + \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_u^{pr} y_u^{pr} LF_{5,1,-2}[m_u^r, m_q^P] \delta_{i1i2} + \\
& \quad \frac{5}{36} g_1^4 LF_{4,1,-2}[\tilde{\mu}, m_1] \delta_{i1i2} - \frac{1}{9} g_1^4 LF_{5,1,-3}[\tilde{\mu}, m_1] \delta_{i1i2} + \frac{5}{12} g_1^2 g_2^2 LF_{4,1,-2}[\tilde{\mu}, m_2] \delta_{i1i2} - \\
& \quad \frac{1}{3} g_1^2 g_2^2 LF_{5,1,-3}[\tilde{\mu}, m_2] \delta_{i1i2} - \frac{1}{2} \overline{y}_d^{ps} \overline{y}_d^{ri1} y_d^{pi2} y_d^{rs} LF_{2,1,0}[\tilde{\mu}, m_d^S] + \\
& \quad \overline{y}_d^{ps} \overline{y}_d^{ri1} y_d^{pi2} y_d^{rs} LF_{3,1,-1}[\tilde{\mu}, m_d^S] - \frac{1}{2} \overline{y}_d^{ps} \overline{y}_d^{ri1} y_d^{pi2} y_d^{rs} LF_{4,1,-2}[\tilde{\mu}, m_d^S] + \\
& \quad \frac{1}{36} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{2,1,0}[\tilde{\mu}, m_q^P] + \frac{7}{36} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{3,1,-1}[\tilde{\mu}, m_q^P] - \\
& \quad \frac{1}{18} g_1^2 \overline{y}_d^{pi1} y_d^{pi2} LF_{4,1,-2}[\tilde{\mu}, m_q^P] - \frac{1}{2} \overline{y}_d^{pi1} y_d^{ri2} \overline{y}_u^{rs} y_u^{ps} LF_{2,1,0}[\tilde{\mu}, m_u^S] + \\
& \quad \$$